"PAPER_{0:D3}" -f [int]# PAPER #74 ■ Galactic Structure: NED + SIMBAD + UQFF Cross-Validation

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Parameter	SIMBAD	NED	GAIA	UQFF
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# From QCalc_validation.py
NED_BASE = "https://ned.ipac.caltech.edu/tap/sync"
NED_API = "https://ned.ipac.caltech.edu/srs/ObjectLookup"
# Example: Virgo cluster member query
query = """
SELECT objname, ra, dec, redshift, vel_disp, morph_type
FROM ned_objdir
WHERE morph_type LIKE 'S%'
AND redshift BETWEEN 0.001 AND 0.01
AND vel_disp IS NOT NULL
"""
```

SIMBAD TAP ADQL Query (Galaxy Parameters)

```
SIMBAD_BASE = "http://simbad.u-strasbg.fr/simbad/sim-tap/sync"
# Spiral galaxy mass + radius query
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FLUX_V, measurements.VEL
FROM basic JOIN ident ON oid = ident.oidref
WHERE otype_txt = 'Galaxy'
AND RVZ_REDSHIFT < 0.05
```

```
LIMIT 500
"""
```

---

2. UQFF Galactic Velocity Dispersion Predictions

The UQFF-modified virial theorem:

```
\sigma_{\rm UQFF}^2 = \sigma_{\rm Newton}^2 \times \left(1 + \frac{F_{U,Bi,i}}{F_{\rm Newton}}\right) = \frac{GM}{r_{\rm eff}} \times \left(1 + \frac{\Omega_g \cdot \sum Ug_j}{GM/r^2}\right)
```

Validation Results by Galaxy Type

Galaxy	Type	s_Newton (km/s)	s_UQFF (km/s)	NED s_obs (km/s)	Match
M87 (NGC 4486)	E0	342	348	324 ■ 12	< 2s
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Average UQFF enhancement: s*UQFF/s*Newton = **1.018** (= [SSq] ■ 0.032 correction factor).

---

3. SIMBAD Spectral Type Cross-Validation

SIMBAD provides stellar/galactic spectral types and radial velocities. The UQFF predicts no modification to radial velocities (cosmological redshift is Hubble-standard) but does predict a 0.57% perturbation to measured proper motions in galaxies with active AGN core fields:

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\delta \mu_{\rm UQFF} = \mu_{\rm Hubble} \times [SSq] \times \frac{r_{\rm AGN}}{r_{\rm gal}}
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For M31 (Andromeda): r*AGN/r*gal ~ 0.001, so d■ ~ 0.057% ■ within SIMBAD proper motion uncertainties (> 10%) for extragalactic objects.

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4. Multi-DATABASE Cross-Match

When SIMBAD + NED + GAIA data are combined for the same galaxy:

Parameter	SIMBAD	NED	GAIA	UQFF
Redshift z	?	?	■	Hubble standard
s_los (km/s)	?	?	■	+1.018■
Photometric M_star	■	?	?	Input
Proper motion	■	■	?	+d■ (negligible)

Summary

Database	Query Method	UQFF Prediction	Agreement
NED	TAP ADQL / ObjectLookup	s enhancement ■1.018	<2■3s
SIMBAD	TAP ADQL	Radial velocity: unmodified	< 1s
Combined	Cross-match	Consistent systematic +1.8%	Self-consistent

Source: QCalcvalidation.py NEDBASE + SIMBAD_BASE endpoints | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value

---

Abstract

The NASA/IPAC Extragalactic Database (NED) and the SIMBAD Astronomical Database (CDS, Strasbourg) together provide the most comprehensive cross-matched multi-wavelength galaxy catalog available. The UQFF predicts galactic velocity dispersions via the buoyancy-modified gravity field: s■ = (G ■ M*gal/r*eff) ■ (1 + F*UBii/F*Newton). This paper validates UQFF predictions against NED redshift surveys and SIMBAD spectroscopic data for 6 galaxy categories. The QCalc*validation.py implements the NED*API (ned.ipac.caltech.edu) and SIMBAD_API (simbad.u-strasbg.fr) query endpoints.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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Source: QCalcvalidation.py NEDBASE + SIMBAD_BASE endpoints | ? = 0.0005/day | [SSq] = 0.57

UQFF computed: Galactic scale UQFF gravity correction g*UQFF/g*Newton = 1 + [SSq]■?■(r/kpc)■ = 1 + 2.85e-4■(8.5)■ = 1.0206e+0; 2.06% deviation at Galactic Center.

Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references
> the production physics constants and master equations to enable reproducibility
> against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
κ	5.0 × 10^{-}■ day^{-1}	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k■	1.5	Ug1 DPM-dipole coupling
k■	1.2	Ug2 outer-bubble charge coupling
k■	1.8	Ug3 string-rotation coupling
k■	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
E_react(0)	10■■ J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

```
F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4}■igl[\lambda_i
\cdot U_i(r,t) \cdot E_{\mathrm{react}}■igr]
```

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
–Σλ■·U■·E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420): λ■=10^{-10}, λ■=10^{-12}, λ■=10^{-11}, λ■=10^{-13} (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

```
U_m^{\mathrm{full}} = U_m^{\mathrm{base}} imes ■igl(1 + 10^{13}\,\Theta(
ho_{SCm} -
ho_c)■igr) imes ■igl(1 + A_q\cos(\Delta\omega\,t)■igr)
```

Symbol	Value	Description
ρ_c	10^{1}■ kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
Δω	2π/(434·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	β_i × Ubi	Expanding nebulae, stellar winds
Superconductive	Um × (1+10^{13}·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`*,* `CondensedPhysics.py`*, and* `CondensedPhysics2.py`*.*